1. Introduction

Large language models (LLMs) (Anthropic, 2023; Brown et al., 2020; Google, 2023; Jiang et al., 2023a; OpenAI, 2022, 2023; Radford et al., 2018, 2019; Gemini Team, 2023; Touvron et al., 2023a,b; Yin et al., 2023; Zhao et al., 2023) based on Transformers (Vaswani et al., 2017) have become a prominent pathway to Artificial General Intelligence (AGI). LLMs acquire massive world knowledge by learning to predict the next word on large-scale web corpora. The capabilities of LLMs have been continuously increasing by scaling model sizes, dataset sizes, and computation. After pre-training, LLMs can be aligned to support real-world use cases by supervised fine-tuning (Chung et al., 2022; Sanh et al., 2022) and preference optimization techniques including reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022a; Wang et al., 2024; Zheng et al., 2023b) and direct preference optimization (DPO) (Rafailov et al., 2023). The capabilities of LLMs have empowered various applications including ChatGPT, Claude, Bard, Microsoft Copilot, Character.AI, Notion AI, etc. Recently, many specialized LLMs have been trained for different targeted usage scenarios. In general, LLMs specialize according to the targeted domains (e.g., finance (Wu et al., 2023), healthcare (Yang et al., 2022b), legal (Cui et al., 2023), etc.) and tasks (e.g., role-playing (Wang et al., 2023d), coding (Rozière et al., 2023), etc.). However, the ability of LLMs to write human-like texts and produce creative content, which is a critical use case of LLM applications such as ChatGPT, is mostly overlooked by the community.

In this report, we focus on the ***literature*** domain and the task of ***writing***, or ***content creation***, and introduce Weaver, a family of LLMs dedicatedly pre-trained and aligned for this purpose. The name "Weaver" symbolizes the model's proficiency in skillfully amalgamating linguistic elements, akin to the way a craftsman weaves threads to form a fabric. We answer four main questions in this technical report: ***why we need Weaver***, ***how we train Weaver***, ***how Weaver performs***, and ***what we build with Weaver***.

Why we need Weaver? Despite generalist LLMs such as GPTs already possessing general writing skills and helping billions of users in various writing scenarios, they often struggle to produce human-like texts in specific writing scenarios such as writing stories, fiction, social media copies, blogs, papers/thesis, etc. We analyze the behavior of pre-trained base LLMs such as LLaMA and aligned LLMs such as ChatGPT and LLaMA-chat and believe this limitation originates from both the pre-training stage and the alignment stage. On one hand, generalist LLMs are pre-trained on massive low-quality web texts or machine/AI-generated texts. Consequently, existing LLM backbones tend to produce seemingly fluent texts that are not creative enough and lack human-like styles. On the other hand, during the alignment stage, state-of-the-art LLMs such as GPT-4 are instruction-tuned using instruction-response pairs annotated by crowdsource annotators (Ji et al., 2023; Shen et al., 2023; Wang et al., 2023c). However, most of the annotators are not professional writers or content creators and the annotation guidelines only require them to produce helpful and harmless responses (Ouyang et al., 2022b). As a result, the crowdsourced data for supervised fine-tuning is less stylish and lacks creativity. Furthermore, most popular preference

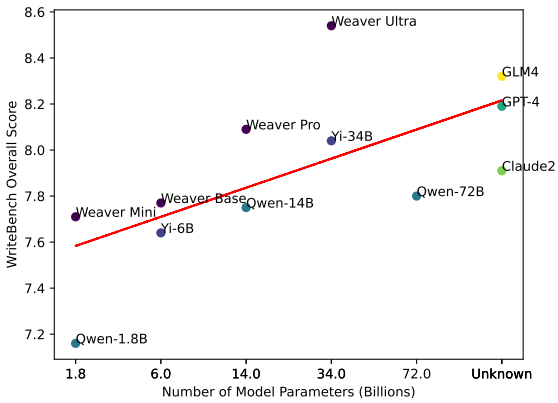

Figure 1 | Comparison between Weaver and generalist LLMs on WriteBench.

optimization methods such as RLHF and DPO optimize the model on model-generated data pairs, making them less suitable for enhancing the creativity of LLMs.

These factors make current generalist LLMs lack creativity and unable to produce human-style texts despite they are super powerful in other applications such as writing codes and answering general questions. We believe this phenomenon will continue to be amplified given that the amount of LLM-generated texts on the internet is exponentially growing and most LLMs are aligned using texts produced by other LLMs. Therefore, we believe it is necessary to train domain-specific LLMs dedicated to writing purposes that are creative and generate human-like texts in order to fully exploit the potential of AI-generated content (AIGC).

How we train Weaver? To address the aforementioned issues limiting generalist LLMs' creative writing ability, we carefully design a suite of strategies for automated data collection, data annotation, and data filtering for pre-training and alignment. This makes us able to pre-train and align Weaver on diverse, human-like, and stylish texts. To be specific, we conduct extensive pre-training data filtering and only keep high-quality content such as books, fiction, stories, and articles in the pre-training corpus, making the pre-trained backbones more likely to produce human-like texts.

As for the alignment stage, we propose a new instruction backtranslation framework inspired by LongForm (Köksal et al., 2023) and Humpback (Li et al., 2023) that synthesize diverse and natural instructions that correspond to high-quality outputs written by professional writers and preferred by human consumers. Our instruction backtranslation framework translated the work of crowdsource annotators from writing both instructions and outputs to simply collecting high-quality content such as stories, fiction, articles, social media copies, and blog posts. This massively reduces the cost of instruction data annotation and the requirement for crowdsource annotators while significantly improving the quality of annotated data.

Moreover, we also propose a novel Constitutional DPO algorithm for preference optimization to better align Weaver to the preference of professional writers and content creators. Constitutional DPO is inspired by and combines the advantages of a few previous works including DPO (Rafailov et al., 2023), Constitutional AI (Bai et al., 2022), Self-Align (Sun et al., 2023), and RLCD (Yang et al., 2023a). Specifically, Constitutional DPO exploits expert (e.g., professional editors in our case) annotated principles to synthesize negative examples that violate certain principles based on positive examples that are sampled from the optimal policy (e.g., texts produced by professional writers or content creators in our case). In contrast to the common practice of using DPO that uses LLMs to produce preference annotation on two model-generated responses such as Zephyr (Tunstall et al., 2023), the pairwise preference data synthesized by our approach contains less noise since the negative example are deliberately synthesized to be of lower quality compared to the positive example. The pairwise preference data generated by Consitutional DPO also contains more principled and targeted learning signals that can be adjusted by human experts according to target domains and applications.

Furthermore, we propose to transform the annotation instructions and responses used in the instruction backtranslation and Constitutional DPO stages into annotation instructions and evaluation instructions. In this way, Weaver not only possesses abilities to follow writing instructions but can also annotate writing instructions and evaluate writing outputs. We also curate instruction data for retrieval-augmented generation (RAG) and function calling to enable Weaver to exploit external knowledge and tools. The combination of different data sources makes Weaver a versatile foundation model while specializing in creative writing.

How Weaver performs? Evaluating the content creation/writing ability of LLMs remains an open problem since existing benchmarks for LLMs such as MMLU (Hendrycks et al., 2020) or MT-Bench (Zheng et al., 2023a) mostly focus on reasoning, math, coding, or general questions instead of

creative writing. Moreover, it is already notoriously hard to evaluate LLMs on general instructions, and it becomes much harder for creative writing tasks since literary critic is non-trivial even for human experts, not to mention LLMs. To better evaluate WEAVER and help the LLM community better measure progress on AIGC, we carefully curate WRITEBENCH, a benchmark for assessing the creative writing capabilities of LLMs and collect outputs from 10+ popular LLMs covering both open-source and proprietary models.

We then conduct both LLM-based and human evaluation of WEAVER and generalist LLMs on the benchmark. Evaluation results confirm the superiority of WEAVER compared to generalist LLMs. We find that WEAVER ULTRA, the most-capable model in the WEAVER family, advances the state-of-the-art in creative writing despite being 10+ smaller compared to GPT-4[1], the previous best performing LLM. Other models in the WEAVER family also surpass competitive generalist LLMs several times larger than them. Our analysis and case studies show that the main source of improvements is because WEAVER can generate texts that are creative and human-like while generalist LLMs tend to produce too "predictable" texts. To confirm that WEAVER is *truly helpful* in real-world applications, we also conduct a user study where human writers are asked to write stories (fiction writing) and blog posts (non-fiction writing) with WEAVER and GPT-4. Our user study shows that compared to GPT-4, WEAVER improves the writers' productivity by 47% and helps writer produce better stories and articles at the same time.

What we build with WEAVER? Training specialized LLMs for writing is one side of enhancing AI-assisted writing experience. We believe it is also very important to build a better human-AI interface to fully exploit the potential of WEAVER on AI-assisted writing. To this end, we introduce WAWAWRITER, our innovative human-AI collaborative writing platform. Similar to recent AI writing products such as Notion AI, WAWAWRITER provides a chat interface that allows users to provide diverse writing instructions, instead of merely suggesting the next one or few sentences based on the current context or polishing the content as in traditional applications. WAWAWRITER also takes a few steps further: (1) we enable ***human-AI co-editing*** by allowing users to customize language agents (Zhou et al., 2023b) that acts like a human collaborator by operating inside the editor simultaneously with users; (2) we allow users to build ***personal knowledge bases*** by saving websites or uploading documents and build a RAG pipeline that integrates knowledge bases to WEAVER; (3) we propose to provide ***personalized writing assistance*** by analyzing users' personal writing styles using LLMs based on their writing history on the platform and using the results to guide WEAVER's text generation process. By combining these innovations, WAWAWRITER aims to provide next-generation AI-assisted writing experience that is more helpful and enjoyable.

In the following sections, we first describe the architectures and sizes of the WEAVER family and their pre-training stage. We then present details on the abilities of WEAVER, how we synthesize training data to help WEAVER acquire these abilities and learn to produce human-like stylish texts, and the details for the alignment stage. We also present our benchmark for evaluating the writing abilities of LLMs and the evaluation results. Finally, we introduce the details of WAWAWRITER and present how WEAVER paves the way for next-generation AI-assisted writing experiences.

2. Pre-training

2.1. Model Family

WEAVER models are language models built on top of Transformer decoders. We have adopted the recent improvements from the design of LLaMA (Touvron et al., 2023a,b), the most popular open-

[1]According to non-official rumor about the size of GPT-4